

Math 427 homework 2

Eric Zhang

Problem 2.1.13

Proof. Let $W \subseteq \mathbb{C}$ be an open set. Then $f^{-1}(W) \subseteq U_f$ is open since f is continuous. Similarly $g^{-1}(f^{-1}(W)) \subseteq U_g$ is open as g is continuous. Now $(f \circ g)^{-1}(W) = g^{-1}(f^{-1}(W))$ is open and it follows from Theorem 2.1.13 that $f \circ g$ is continuous. $\square$

Problem 2.1.14

Proof. Assume $f : U \rightarrow \mathbb{C}$ is continuous. Note that $g : \mathbb{C} \rightarrow \mathbb{C}$ defined by $z \mapsto |z|$ is also continuous by Example 2.1.2. Then $f(U) \subseteq U_g = \mathbb{C}$ and hence $g \circ f$ is continuous by Problem 2.1.13. It follows that $\{z \in U : |f(z)| < r\} = (g \circ f)^{-1}(D_r(0))$ is open. $\square$

Proof. To see $\{z \in U : \operatorname{Re}(f(z)) < r\}$ is open, it suffices to show $g : \mathbb{C} \rightarrow \mathbb{C}$ defined by $z \mapsto \operatorname{Re}(z)$ is continuous since $\{z \in U : \operatorname{Re}(f(z)) < r\} = (g \circ f)^{-1}(D_r(0))$. Note that for any $\epsilon > 0$, choose $\delta = \epsilon$, then $|z - w| < \delta$ implies $|\operatorname{Re}(z) - \operatorname{Re}(w)| \leq |z - w| < \epsilon$. Or we can also note that $\{z_n\}$ converges implies $\{\operatorname{Re}(z_n)\}$ converges as well (and continuity follows from Problem 2.1.16). $\square$

Problem 2.3.16

Proof. ($\implies$) Let f be continuous. Assume $\{z_n\}$ converges to a where z_n and $a \in U$. Then

($\impliedby$) Suppose for any sequence $\{z_n\}$ converges to a where z_n and $a \in U$, the corresponding sequence $\{f(z_n)\}$ converges to $f(a)$. $\square$

Problem 1.4.3

Proof.

$\square$

Problem 1.4.9*Proof.*

□

Problem 1.4.18*Proof.*

□